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Integrated Circuit Structure and Method with Hybrid Orientation for FinFET

Abstract

The present disclosure provides a semiconductor structure. The semiconductor structure includes a semiconductor substrate having a first region and a second region; a first fin active region of a first semiconductor material disposed within the first region, oriented in a first direction, wherein the first fin active region has a <100> crystalline direction along the first direction; and a second fin active region of a second semiconductor material disposed within the second region and oriented in the first direction, wherein the second fin active region has a <110> crystalline direction along the first direction.

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Background/Summary

BACKGROUND [0001] This application is a Continuation of U.S. patent application Ser. No. 18/355,895, filed Jul. 20, 2023, which is a Continuation of U.S. patent application Ser. No. 17/341,142, filed Jun. 7, 2021 (now U.S. Pat. No. 11,735,594), which is a Continuation of U.S. patent application Ser. No. 16/741,530, filed Jan. 13, 2020 (now U.S. Pat. No. 11,031,418), which is a Divisional of U.S. patent application Ser. No. 15/800,390, filed Nov. 1, 2017 (now U.S. Pat. No. 10,535,680), which further claims priority to U.S. Provisional Patent Application Ser. No. 62/526,471, entitled “Integrated Circuit Structure and Method with Hybrid Orientation for FinFET,” filed on Jun. 29, 2017, the entire disclosures of which are hereby incorporated herein by reference.

[0002] The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs.

[0003] Such scaling down has also increased the complexity of processing and manufacturing ICs and, for these advances to be realized. Similar developments in IC processing and manufacturing are needed. For example, a three-dimensional transistor, such as a fin-like field-effect transistor (FinFET), has been introduced to replace a planar transistor. Furthermore, the carrier mobility is desired to increase for device speed and performance. However, the existing structures and associated methods are not proper to the 3D structure including FinFETs.

[0004] Although existing FinFET devices and methods of fabricating FinFET devices have been generally adequate for their intended purposes, they have not been entirely satisfactory in all respects. For example, the existing structures and associated methods are improper or not optimized to the 3D structure including FinFETs, especially for the high mobility channel.

Therefore, what are needed are an integrated circuit structure and the method making the same to address the above issues.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0006] FIG. 1 is a flowchart of an integrated circuit fabrication method in accordance with some embodiments.

[0007] FIG. 2 is a top view of two semiconductor substrates to be bonded to form a semiconductor structure, constructed in accordance with some embodiments.

[0008] FIG. 3 is a perspective view of the two semiconductor substrates to be bonded to form the semiconductor structure, constructed in accordance with some embodiments.

[0009] FIG. 4 is a top view of the semiconductor structure at a fabrication stage, constructed in accordance with some embodiments.

[0010] FIGS. 5, 6, 7, 8, 9, 10 and 11A are sectional views of the semiconductor structure at various fabrication stages, constructed in accordance with some embodiments.

[0011] FIG. 11B is a sectional view of the semiconductor structure constructed in accordance with other embodiments.

[0012] FIG. 12A is a top view of the semiconductor structure, in portion, in accordance with some embodiments.

[0013] FIG. 12B is a perspective view of the semiconductor structure, in portion, in accordance with some embodiments.

[0014] FIGS. 13A and 13B are diagrams illustrating experimental data of the semiconductor structure, constructed in accordance to some examples.

DETAILED DESCRIPTION

[0015] The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0016] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as being “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the exemplary term “below” can encompass both an orientation of above and below. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted

accordingly.

[0017] The present disclosure is directed to, but not otherwise limited to, a fin-like field-effect transistor (FinFET) device. The FinFET device, for example, may be a complementary metal-oxide-semiconductor (CMOS) device including a P-type metal-oxide-semiconductor (PMOS) FinFET device and an N-type metal-oxide-semiconductor (NMOS) FinFET device. The following disclosure will continue with a FinFET example to illustrate various embodiments of the present invention. It is understood, however, that the application should not be limited to a particular type of device, except as specifically claimed.

[0018] FIG. 1 is a flowchart of a method **100** for fabricating a semiconductor structure (particularly including a FinFET structure having one or more FinFET devices) **200**, constructed in accordance with some embodiments. In accordance with some embodiments, FIG. 2 is a top view of the two semiconductor substrate to be bonded to form the semiconductor structure **200**; FIG. 3 is a perspective view of the two semiconductor substrates to be bonded. FIG. 4 is a top view of the semiconductor structure **200** at a fabrication stage. FIGS. 5 through 11 are sectional views of the semiconductor structure **200** at various fabrication stages. FIG. 12A is a top view of the semiconductor structure **200**, in portion. FIG. 12B is a perspective view of the semiconductor structure **200**, in portion. FIGS. 13A and 13B are diagrams illustrating experimental data of the semiconductor structure **200** according to some examples. It is understood that additional steps may be implemented before, during, and after the method, and some of the steps described may be replaced or eliminated for other embodiments of the method. The semiconductor structure **200** and the method **100** making the same are collectively described with reference to FIGS. 1 through 13B.

[0019] The method **100** begins with two semiconductor substrates **202** and **204**, as illustrated in FIG. 2 in a top view. Particularly, the first semiconductor substrate **202** and the second semiconductor substrate **204** are the same in terms of crystalline structure and plane orientation. In the present embodiment, the first semiconductor substrate **202** and the second semiconductor substrate **204** are both silicon substrates, such as silicon wafers. In furtherance of the present embodiment, the first semiconductor substrate **202** and the second semiconductor substrate **204** are silicon wafers in crystalline structure with a same plane orientation, such as (100) in the present example. Here (xyz) is a miller index representing a plane orientation of the top surface of the crystalline silicon wafer. Accordingly, the crystal directions $\langle 100 \rangle$ of the semiconductor substrates **202** and **204** are oriented in the top surface of the respective substrate and are labeled as **206** and **208**, respectively, as illustrated in FIG. 2. Here $\langle 100 \rangle$ is another miller index representing a family of crystal directions of a crystalline semiconductor substrate.

[0020] Even though the two semiconductor substrates are silicon substrates in the present embodiment. However, the disclosed structure and the method are not limiting and are extendable to other suitable semiconductor substrates and other suitable orientations. For examples, the substrate **210** may include an elementary semiconductor, such as germanium in a crystalline structure; a compound semiconductor, such as silicon germanium, silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; or combinations thereof. In furtherance of the embodiments, those semiconductor material films may be epitaxially grown on the silicon wafer.

[0021] Referring to FIG. 1, the method **100** includes an operation **102** to bond the two semiconductor substrates **202** and **204** with an orientation offset. In the present embodiment, the two substrates **202** and **204** are bonded together in a way such that a same crystal direction (such as $\langle 100 \rangle$) of the two substrates **202** and **204** has a 45° degree angle. The operation **102** is further described below in details.

[0022] The first substrate **202** and the second substrate **204** are rotated and configured so that the corresponding $\langle 100 \rangle$ crystal directions are offset with an angle **302** therebetween, as illustrated in FIG. 3. The angle **302** is 45° degree. In the present embodiment, the top surface **304** of the first substrate **202** and the top surface **306** of the second substrate **204** are configured in a way that both

face a same direction, as illustrated in FIG. 3.

[0023] The two substrates **202** and **204** are bonded together with such configuration through a proper bonding technology, such as direct bonding, eutectic bonding, fusion bonding, diffusion bonding, anodic bonding or other suitable bonding method. In one embodiment, the substrates are bonded together by direct silicon bonding (DSB). For example, the direct silicon bonding process may include preprocessing, pre-bonding at a lower temperature and annealing at a higher temperature. A buried silicon oxide layer (BOX) may be implemented when the two substrates are bonded together. In some examples, a thermal silicon oxide film is formed on one bonding surface (or both bonding surfaces: the top surface of the first substrate **202** and the bottom surface of the second substrate **204**) by thermal oxidation. Then the bonding surfaces are positioned with the orientation and configuration as described above in FIG. 3. The bonding surfaces are brought together and bonded by thermal heating and mechanical pressure. The thermal heating may have a temperature at or greater than 1000° C. The thermal heating and mechanical pressure may have different profiles during the bonding process to optimize the bonding effect. For example, the bonding process may include two steps: the first step at room temperature with a mechanical pressure and the second step with a thermal annealing greater than 1000° C. In some examples, at least one of the semiconductor substrates **202** and **204** may be thinned down, such as by grinding or polishing, to proper thicknesses before or after the bonding. In an alternative embodiment, the buried oxide layer may be eliminated in a way that the silicon surfaces are directly bonded together. In another embodiment, the thermal annealing temperature may be reduced to a relatively lower temperature, such as a temperature between 400° C. and 450° C.

[0024] The bonded two substrates are illustrated in FIG. 4 in a top view. X and Y directions are Cartesian coordinate and defined on the (100) plane surface. The <100> crystal direction **208** of the second substrate **204** is along the Y axis while the <100> crystal direction **206** of the first substrate **202** is along a diagonal direction between X and Y axes with the angle **302** of 45 degree from the Y axis. Particularly, the first substrate **202** has the crystal direction <110> along the Y axis.

[0025] FIG. 5 is a sectional view of the semiconductor structure **200** having the first substrate **202** and the second substrate **204** bonded together. In the present embodiment, the silicon oxide film **502** is interposed between the two substrates. Again, the top surfaces of the substrates are (100) oriented but the first substrate **202** has a rotation of 45° to the second substrate **204** so that the first substrate **202** has the crystal direction <110> along the Y axis and the second substrate **204** has the crystal direction <100> along the Y axis.

[0026] Referring to FIGS. 1 and 6, the method **100** proceeds to an operation **104** to form a patterned mask on the second substrate **204** of the semiconductor structure **200**. The semiconductor structure **200** includes a first region **602** for an n-type field-effect transistor (NFET) and a second region **604** for a p-type field-effect transistor (PFET). The patterned mask covers the first region **602** and includes an opening that exposes the second region **604**. The patterned mask may be a soft mask (such as a patterned resist layer) or a hard mask (such as a dielectric material layer). In the present embodiment, the hard mask is used. A hard mask **608** is disposed on the top surface of the second substrate **204**; a patterned resist layer **610** is formed on the hard mask **608** by a lithography process; and the hard mask **608** is etched to transfer the opening from the patterned resist layer **610** to the hard mask **608**. In some examples, the hard mask **608** includes a silicon oxide and a silicon nitride subsequently deposited on the semiconductor structure **200**. The hard mask **608** may be formed by thermal oxidation, chemical vapor deposition (CVD), atomic layer deposition (ALD), any other appropriate method, or a combination thereof. An exemplary photolithography process may include forming a resist layer, exposing the resist by a lithography exposure process, performing a post-exposure bake process, and developing the photoresist layer to form the patterned photoresist layer. The lithography process may be alternatively replaced by other technique, such as e-beam writing, ion-beam writing, maskless patterning or molecular printing. In some other embodiments, the patterned resist layer may be directly used as an etch mask for the

subsequent etch process. The patterned resist layer **610** may be removed by a suitable process, such as wet stripping or plasma ashing, after the formation of the hard mask **608**.

[0027] Referring to FIGS. **1** and **7**, the method **100** includes an operation **106** to etch the second semiconductor substrate **204** in the second region **604** until the first substrate **202** is exposed within the second region **604**, resulting in a trench **702**. The etching process is designed to selectively remove the semiconductor material(s) in the second region **604** using the hard mask **608** as an etch mask. The etching process may further continue to recess the first semiconductor substrate **202** or ensure the first semiconductor substrate **202** within the second region **604** is exposed. The etching process may include dry etch, wet etch or a combination thereof. The patterned mask **608** protects the second substrate **204** within the first region **602** from etching. In various examples, the etching process includes a dry etch with a suitable etchant, such as fluorine-containing etching gas or chlorine-containing etching gas, such as Cl.sub.2, CCl.sub.2F.sub.2, CF.sub.4, SF.sub.6, NF.sub.3, CH.sub.2F.sub.2 or other suitable etching gas. In some other examples, the etching process includes a wet etch with a suitable etchant, such as KOH solution. The etching process may include more than one step. For example, the etching process may include a first etching step to etch the silicon material of the second substrate **204** and a second etching step to etch the silicon oxide layer **502**. In furtherance of the example, the etching process includes a dry etch step using fluorine-containing etching gas or chlorine-containing etching gas to etch silicon and a wet etch step using hydrofluoric acid solution to etch silicon oxide. In some embodiments, a spacer **704** may be formed on sidewalls of the trench **702**. The spacer **704** may include a dielectric material to on the sidewall of the trench **702** so that the subsequent epitaxial grown semiconductor material in the trench **702** can carry the crystal orientation from the first semiconductor substrate **202** without being impacted by the second substrate **204**. Furthermore, the spacer **704** provides isolation between the first region **602** and the second region **604**. The spacer **704** may be formed by a procedure that includes deposition (such as CVD or PVD) and anisotropic etching, such as dry etching. In some examples, the spacer **704** includes silicon nitride, silicon oxide, titanium nitride or a combination thereof.

[0028] Referring to FIGS. **1** and **8**, the method **100** proceeds to an operation **108** to epitaxially grow a semiconductor material **802** from the first semiconductor substrate **202** within the second region **604**. The semiconductor material **802** fills in the trench **702**. In one embodiment, the semiconductor material **802** is different from the semiconductor material of the second semiconductor substrate **204** to achieve strained effect for enhanced mobility. For example, the second semiconductor material **802** is silicon germanium (SiGe). In other examples, the second semiconductor material **802** may be silicon (Si), germanium (Ge) or other suitable elementary semiconductor material or compound semiconductor material. In some embodiment, the epitaxy growth in the operation **108** may include more than one step to grow multiple semiconductor layers with different semiconductor materials for enhanced device performance and other considerations. In the present embodiment, the operation **108** includes a first epitaxial growth of silicon to form a silicon layer **804** and a second epitaxial growth of SiGe to form a SiGe layer **802**.

[0029] Referring to FIGS. **1** and **9**, the method **100** may proceed to an operation **110** to perform a polishing process, such as chemical mechanical polishing (CMP) to planarize the top surface. The hard mask **608** may be used as a polishing stop layer during the CMP process and is removed by etching after the CMP. Alternatively, the hard mask **608** may be removed by the CMP process.

[0030] The semiconductor layer **802** vertically grows from the semiconductor layer **804** and substantially fills in the trench **702**. In the present embodiment, there may be a gap between the semiconductor layer **802** and the spacer **704**, due to the presence of the spacer **704**. In this case, the polishing process in the operation **110** additionally eliminates or reduces such gap.

[0031] Since it is formed by epitaxial growth from the first semiconductor substrate **202** that has a rotation relative to the second semiconductor substrate **204**, the semiconductor material **802** is in the crystalline structure and carries the same crystal orientations of the first semiconductor substrate **202**. For example, the top surface of the second semiconductor material **802** within the

second region **604** is still in the (100) crystal plane while the top surface of the second semiconductor substrate **204** within the first region **602** has the (100) crystal plane as well. However, the crystal directions for those two regions are different, due to the rotation. Particularly, the second semiconductor substrate **204** within the first region **602** has $\langle 100 \rangle$ crystal directions along both X and Y axes while the second semiconductor material **802** within the second region **604** has $\langle 100 \rangle$ crystal directions along the diagonal directions between X and Y axes. Furthermore, the second semiconductor material **802** within the second region **604** has crystal directions $\langle 110 \rangle$ oriented along the X and Y axes.

[0032] Referring to FIGS. **1** and **10**, the method **100** may proceed to an operation **112** to form fin-type active regions **1002** within the first region **602** and fin-type active regions **1004** within the second region **604**. The fin-type active regions (or simply referred to as fins or fin features) are three-dimensional (3D) semiconductor features where various devices, such as FETs, are formed thereon.

[0033] In some embodiments, the fins **1002** and **1004** are formed by any suitable procedure including deposition, photolithography, and/or etching processes. In one embodiment, the fins **1002** and **1004** are formed by patterning the semiconductor structure **200** (the second substrate **204** in the first region **602** and the semiconductor material **802** in the second region **604**) to form trenches. In furtherance of the embodiment, the patterning of the semiconductor structure **200** may include forming a hard mask; and applying an etch process to the semiconductor structure **200** through the openings of the hard mask to form trenches in the semiconductor structure **200**. The hard mask may be formed by a procedure that includes depositing a hard mask material layer and etching the hard mask material layer. In some examples, the hard mask material layer includes a silicon oxide layer and a silicon nitride layer subsequently deposited on the semiconductor structure **200**. The hard mask material layer may be formed by thermal oxidation, chemical vapor deposition (CVD), atomic layer deposition (ALD), or any other appropriate method. The procedure to form the patterned hard mask further includes forming a patterned photoresist (resist) layer by a lithography process, and etching the hard mask material layer through the openings of the patterned resist layer to transfer the openings to the hard mask material layer. In some other embodiments, the patterned resist layer may be directly used as an etch mask for the etch process to form the trenches. The etching process may include more than one step and may continue to etch through the silicon oxide layer **502** within the first region **602** and the silicon layer **802** within the second region **604** to reach the first semiconductor substrate **202**, as illustrated in FIG. **10**.

[0034] The fins **1002** within the first region **602** may be referred to as n-type fins because the semiconductor material therein is doped appropriately to form NFETs. Specifically, the semiconductor material is doped with a p-type dopant such as boron. Similarly, the fins **1004** within the second region **604** may be referred to as p-type fin features because they will ultimately be doped appropriately to form a p-type transistor. Specifically, they will be doped with an n-type dopant such as phosphorous. The dopants may be introduced into the respective regions by ion implantation or in-situ doping.

[0035] The fins have elongated shape and are oriented in the Y direction. Particularly, the n-type fins **1002** have crystal direction $\langle 100 \rangle$ in the Y direction while the p-type fins **1004** have crystal direction $\langle 110 \rangle$ in the Y direction, due to the rotation described above. Furthermore, both X and Y directions of the n-type fins **1002** have crystal direction $\langle 100 \rangle$ and both X and Y directions of the p-type fins **1004** have crystal direction $\langle 110 \rangle$. Additionally, the sidewall surface orientations of the n-type fin **1002** and the p-type fins are (001) and (110), respectively, as illustrated in FIG. **10**.

[0036] Referring to FIGS. **1** and **11A**, the method **100** may proceed to an operation **116** to form shallow trench isolation (STI) features **1102** to isolate various fin-type active regions. The formation of the STI features **1102** may further includes a first operation **122** to fill in the trenches between fins with one or more dielectric material; a second operation **124** to polish the semiconductor structure **200** for removing excessive dielectric material and planarizing the top

surface; and a third operation **126** to recess the STI features **1102** by selective etching. The second operation **124** to fill in the trenches with the dielectric material(s) may include one or more steps to fill in with respective dielectric materials. For example, a thermal silicon oxide layer is formed in the trenches with a thermal oxidation process and another dielectric material, such as silicon oxide, fills in the trenches by a suitable technique that includes CVD, such as high density plasma CVD (HDPCVD) or flowable CVD (FCVD). The third operation **126** to recess the dielectric material may include an etching process (such as wet etching, dry etching a combination thereof) to selectively etch back the dielectric material, resulting in the STI features **1102**.

[0037] Alternatively, the semiconductor structure **200** may have a different structure, such as one illustrated in FIG. **11B**. In the semiconductor structure **200** of FIG. **11B**, the fin active region **1004** is replaced by a plurality of stacked nanowires **1106**, accordingly a plurality of channels defined in the nanowires **1106**. The formation of the nanowires **1106** is further described here according to some examples. In the operation **108**, the epitaxy growth includes epitaxially growing alternative semiconductor materials, such as Si/SiGe/Si/SiGe. In operation **112**, the formation of the fin active regions includes patterning of the semiconductor materials (as described above) and further includes a selective etching process to selectively remove only one semiconductor material while another remains, such as selectively removing Si (or selectively removing SiGe in an alternative example), thereby forming the nanowires **1106**. Thus, various devices, such as FETs, may be formed on the fins **1002/1004** of FIG. **11A** or alternatively may be formed on the fins **1002** and nanowires **1106** of FIG. **11B**.

[0038] Referring back to FIG. **1**, the method **100** may proceed to an operation **118** to form NFETs on the n-type fins **1002** within the first region **602** and PFETs on the p-type fins **1004** (or nanowires **1106**) within the second region **604**. Alternatively, various semiconductor devices, such as FETs, dipole transistors, diodes, passive devices (resistors, inductors, capacitors or a combination thereof), or a combination thereof are formed on the fins. In the present embodiment illustrated in FIGS. **12A** and **12B**, exemplary NFET **1202** and PFET **1204** are formed on the fins **1002** and **1004**, respectively, and therefore are referred to as fin FETs (FinFETs). FIG. **12A** is a top view and FIG. **12B** is a schematic view of the semiconductor structure **200** in accordance with some embodiments. In FIGS. **12A** and **12B**, only one n-type fin FET (NFinFET) **1202** within the first region **602** and one p-type fin PFET (PFinFET) **1204** within the second region **604** are provided for illustration.

[0039] Specifically, the NFinFET **1202** is formed on the fin **1002** within the first region **602**. The NFinFET **1202** includes source **1206** and drain **1208** formed on the fin **1002**, and a gate **1212** interposed between the source **1206** and drain **1208**. Similarly, the PFinFET **1204** is formed on the fin **1004** within the second region **604**. The PFinFET **1204** includes source **1222** and drain **1224** on the fin **1004**, and a gate **1226** interposed between the source **1222** and drain **1224**.

[0040] Particularly, the channel for each FinFET is defined in the portion of the corresponding fin interposed between the source and drain, and underlying the gate. In the present embodiment, the NFinFET **1202** has a first channel **1232** in the fin **1202** and the PFinFET **1204** has a second channel **1234** in the fin **1204**. For the NFinFET **1202**, the carriers (electrons) flow through the channel **1232** from the source **1206** to the drain **1208** along the direction **1236** that is the crystal direction $\langle 100 \rangle$. For the PFinFET **1204**, the carriers (holes) flow through the channel **1234** from the source **1222** to the drain **1224** along the direction **1238** that is the crystal direction $\langle 110 \rangle$. By providing the semiconductor structure **200** having NFinFETs and PFinFETs with respective crystal directions for channels, the carrier mobility and device performance are enhanced.

[0041] The structure and formation of the FinFETs are further described below. The gate **1212** in the first region **602** includes a gate dielectric feature disposed on the channel and a gate electrode disposed on the gate dielectric feature. The gate may further include gate spacers disposed on sidewalls of the gate electrode. The gate dielectric feature includes a gate dielectric material, such as silicon oxide or a suitable dielectric material having a higher dielectric constant (high-k

dielectric material). In the present embodiment, the gate dielectric feature includes more than one dielectric material layers. For example, the gate dielectric feature includes an interfacial dielectric layer, such as silicon oxide, and a high-k dielectric material layer on the interfacial layer.

[0042] The gate electrode includes a conductive material layer, such as doped polysilicon, metal, metal alloy, metal silicide, or a combination thereof. In some embodiments, the gate electrode includes more than one conductive material layers. For example, the gate electrode includes a first conductive layer having a suitable work function on the gate dielectric feature and a second conductive layer on the first conductive layer. In one example for the n-type FinFET, the first conductive layer includes titanium or titanium aluminum. In another example, the second conductive layer includes aluminum, tungsten, copper, doped polysilicon or a combination thereof.

[0043] Similarly, the gate **1226** in the second region **604** also includes a gate dielectric layer and a gate electrode. The gate electrode in the second region **604** may have a similar structure as that of the gate **1212** in the first region **602**. However, the conductive material(s) may be different. For examples, in the second region **604**, the gate electrode includes a first conductive material layer (such as tantalum nitride or titanium nitride), and a second conductive material layer (such as aluminum, tungsten, copper, doped polysilicon or a combination thereof). The first conductive layers for NFET and PFET are designed with different work functions for optimized threshold voltages.

[0044] The gate (**1212** or **1226**) is formed by a procedure that includes various deposition processes and patterning. The formation of the gate is further described in accordance with some embodiments. In one embodiment, an interfacial layer is formed on the semiconductor substrate (particularly, on the channel). The interfacial layer may include silicon oxide formed by a proper technique, such as an atomic layer deposition (ALD), thermal oxidation or UV-Ozone Oxidation. The interfacial layer may have a thickness less than 10 angstrom. A high k dielectric material layer is formed on the interfacial layer. The high-k dielectric layer includes a dielectric material having the dielectric constant higher than that of thermal silicon oxide, about 3.9. The high k dielectric material layer is formed by a suitable process such as ALD or other suitable technique. Other methods to form the high k dielectric material layer include metal organic chemical vapor deposition (MOCVD), physical vapor deposition (PVD), UV-Ozone Oxidation or molecular beam epitaxy (MBE). In one embodiment, the high k dielectric material includes HfO₂. Alternatively, the high k dielectric material layer includes metal nitrides, metal silicates or other metal oxides. The interfacial layer and the high k dielectric material layer constitute the gate dielectric layer.

[0045] In some embodiments, the gate electrode includes polysilicon. A polysilicon layer is formed on the gate dielectric layer by a proper technique, such as CVD. In one example, a capping layer may be further formed between the high k dielectric material layer and the polysilicon layer by a proper technique, such as PVD. The capping layer may include titanium nitride (TiN), tantalum nitride (TaN) or a combination thereof in some examples. The capping layer may serve one or more functions, such as diffusion barrier, etch stop, and/or protection.

[0046] After the depositions, the gate material layers are patterned to form the gate stack (or a plurality of gate stacks). The patterning of the gate stack includes a lithography process and etching. A lithography process forms a patterned resist layer. In one example, the lithography process includes resist coating, soft baking, exposing, post-exposure baking (PEB), developing, and hard baking. The gate stack material layers are thereafter patterned by etching using the patterned resist layer as an etching mask. The etching process may include one or more etching steps. For example, multiple etching steps with different etchants may be applied to etch respective gate stack material layers.

[0047] In other embodiments, the patterning of the gate stack material layers may alternatively use a hard mask as an etching mask. The hard mask may include silicon nitride, silicon oxynitride, silicon oxide, other suitable material, or a combination thereof. A hard mask layer is deposited on

the gate stack material layers. A patterned resist layer is formed on the hard mask layer by a lithography process. Then, the hard mask is etched through the opening of the patterned resist layer, thereby forming a patterned hard mask. The patterned resist layer may be removed thereafter using a suitable process, such as wet stripping or plasma ashing.

[0048] The gate spacers include a dielectric material and may have one or more films. In some embodiments, the gate spacers include silicon oxide, silicon nitride, silicon oxynitride, other suitable dielectric material, or a combination thereof. The gate spacers are formed by deposition and anisotropic etch (such as dry etch).

[0049] In some embodiments, the source and drain features may further include light-doped drain (LDD) features formed in the fin feature by a suitable process, such as ion implantation. The LDD features have a doping type opposite to that of the channel. For example for an nFET, the channel is p-type and the LDD features are n-type. In another example for a pFET, the channel is n-type and the LDD features are p-type. The LDD features and the gate stack may be formed in a collective procedure. For examples, the gate stack material layers are deposited and patterned to form the gate dielectric layer and gate electrode layer; the LDD features are formed by ion implantation using the dummy gate stack (and STI features) to constrain the LDD features; and then the spacers are formed. In other embodiments, the LDD features are optional and may be eliminated from the semiconductor structure **200**.

[0050] In some alternative embodiments, the gate stack may be formed by a gate replacement process. The gate replacement process include forming a dummy gate; forming source and drain features; removing the dummy gate; and thereafter forming a final gate that includes a gate dielectric layer of a high k dielectric material and a metal gate electrode. In various embodiments, the gate replacement process may be a high-k-last process, in which both the gate dielectric layer and the gate electrode in the dummy gate are removed and replaced by the final gate stack.

[0051] In some alternative embodiments, the source and drain features have different semiconductor material for strained effect and may be formed by a procedure that includes recessing the source and drain regions by etching; and filling the recesses with a different semiconductor (different from the channel material) by epitaxial growth. In some embodiments, the semiconductor material is chosen for proper strained effect in the channel region such that the corresponding carrier mobility is increased. For example, the semiconductor material for the source and drain features includes silicon carbide or silicon germanium.

[0052] The source and drain features may be formed by one or more ion implantations. The following description using the NMOS transistors as an example, and the formation of the source and drain features of PMOS transistors is similar. In some embodiments, the source and drain features for NMOS transistors are formed by an ion implantation procedure that includes a first ion implantation to form LDD features and a second ion implantation to form S/D features. In other embodiments, the formation of the gate stacks and the formation of the source and drain features are integrated in a collective procedure. In furtherance of the embodiments, the gate stacks are deposited and patterned; the first ion implantation is applied to the fin active regions to form LDD features; gate spacers are formed on sidewalls of the gate stacks by a suitable process (such as dielectric deposition and dry etching); and the second ion implantation is applied to the fin active regions to form S/D features.

[0053] The method **100** further includes other operations before, during and/or after the operations described above. For example, the method **100** further includes an operation to form an interlayer dielectric material (ILD) on the semiconductor structure **200**. The ILD includes one or more dielectric materials, such as silicon oxide, a low k dielectric material or a combination thereof. In some embodiment, the formation of the ILD includes deposition and CMP.

[0054] The method **100** includes an operation to form a multilayer interconnection structure to couple various devices to form a functional circuit. The multilayer interconnection structure includes vertical interconnects, such as via features or contact features, and horizontal

interconnects, such as metal lines. The various interconnection features may implement various conductive materials including copper, tungsten, and/or silicide. In one example, a damascene and/or dual damascene process is used to form a copper related multilayer interconnection structure.

[0055] The present disclosure provides a semiconductor structure and a method making the same. The semiconductor structure includes first fin-type active regions for NFinFETs and second fin-type active regions for PFinFETs. The first fin-type active region and second fin-type active regions have channels with different crystal directions, particularly, the channel of the NFinFET is along the crystal direction $\langle 100 \rangle$ and the channel of the PFinFET is along the crystal direction $\langle 110 \rangle$ for enhanced mobility and device performance. The method includes bonding two semiconductor substrates with a rotation such that the above crystal directions are achieved. Furthermore, the channel for PFinFET is also strained for the enhanced mobility by using silicon germanium for the channel.

[0056] The embodiments of the present disclosure offer advantages over existing art, though it is understood that other embodiments may offer different advantages, not all advantages are necessarily discussed herein, and that no particular advantage is required for all embodiments. By utilizing the disclosed method and structure, both the electron mobility for NFinFETs and the electron mobility for PFinFETs are enhanced as illustrated in FIGS. 13A and 13B according to some examples. FIG. 13A illustrates experimental data 1302 of an NFinFET 1202 for its electron mobility versus the width of the corresponding fin 1202. FIG. 13B illustrates experimental data 1304 of a PFinFET 1204 for its hole mobility versus the width of the corresponding fin 1204. The Data 1306 is reference data from an NFinFET with the channel along the $\langle 110 \rangle$ crystal direction and the Data 1308 is reference data from a PFinFET without strain from silicon germanium. From above experimental data, it is clearly show both the mobility for NFinFET 1202 and PFinFET 1204 are enhanced. Especially, the fins may be patterned to produce a relatively close spacing between features, for which the above disclosure is well suited. In addition, spacers used in forming fins of FinFETs, also referred to as mandrels, can be processed according to the above disclosure. In contrary, the existing method and structure for planar devices are not capable of providing combination to optimize and enhance both NFETs and PFETs.

[0057] Thus, the present disclosure provides a semiconductor structure in accordance with some embodiments. The semiconductor structure includes a semiconductor substrate having a first region and a second region; a first fin active region of a first semiconductor material disposed within the first region, oriented in a first direction, wherein the first fin active region has a $\langle 100 \rangle$ crystalline direction along the first direction; and a second fin active region of a second semiconductor material disposed within the second region and oriented in the first direction, wherein the second fin active region has a $\langle 110 \rangle$ crystalline direction along the first direction.

[0058] The present disclosure also provides a method making a semiconductor structure in accordance with some embodiments. The method includes bonding a first semiconductor substrate and a second semiconductor substrate such that corresponding crystalline orientations $\langle 001 \rangle$ having an 45 degree rotation; forming a patterned mask on the second semiconductor substrate to cover a first region and expose a second region; etching the second semiconductor substrate within the second region to expose the first semiconductor substrate; epitaxially growing a second semiconductor material on the first semiconductor substrate within the second region; and patterning the second semiconductor substrate and the second semiconductor material to form a first fin active region in the first region and a second fin active region in the second region. The first fin active region is oriented in a first direction and has a crystalline orientation $\langle 100 \rangle$ along the first direction. The second fin active region is oriented in the first direction and has a crystalline orientation $\langle 110 \rangle$ along the first direction.

[0059] The present disclosure provides a semiconductor structure in accordance with some embodiments. The semiconductor structure includes a semiconductor substrate having a first region

and a second region; a first fin active region of a first semiconductor material disposed within the first region, oriented in a first direction, and being isolated from the semiconductor substrate by a dielectric feature, wherein the first fin active region has a <100> crystalline direction along the first direction; a second fin active region of a second semiconductor material disposed within the second region and oriented in the first direction, wherein the second fin active region has a <110> crystalline direction along the first direction; an n-type fin field-effect transistor (NFinFET) formed on the first fin active region; and a p-type fin field-effect transistor (PFinFET) formed on the second fin active region.

[0060] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor structure, comprising: a semiconductor substrate having a first area and a second area; a first active region of a first semiconductor material disposed within the first area, wherein the first active region has a first crystalline direction along a first direction; and a second active region of a second semiconductor material disposed within the second area, wherein the second active region has a second crystalline direction along the first direction, wherein the second crystalline direction is different from the first crystalline direction, and wherein the second active region includes a plurality of channels vertically stacked and vertically spaced away from each other.
2. The semiconductor structure of claim 1, wherein the second semiconductor material is different from the first semiconductor material in composition; and the first crystalline direction is a <100> crystalline direction, and the second crystalline direction is a <110> crystalline direction.
3. The semiconductor structure of claim 2, wherein each of the first active region and the second active region is in a crystalline structure with a top surface on a (100) crystal plane; the semiconductor substrate is a silicon substrate having a top surface on a (100) crystal plane; and the semiconductor substrate has a crystalline orientation <110> along the first direction.
4. The semiconductor structure of claim 1, further comprising: an n-type field-effect transistor (nFET) formed on the first active region within the first area; and a p-type field-effect transistor (pFET) formed on the second active region within the second area.
5. The semiconductor structure of claim 1, further comprising: a dielectric feature configured to isolate the first active region from the semiconductor substrate; and a semiconductor layer of the first semiconductor material underlying the second active region.
6. The semiconductor structure of claim 5, wherein the dielectric feature spans between opposite edges of the first active region; and the semiconductor layer spans between opposite edges of the second active region.
7. The semiconductor structure of claim 5, wherein the first semiconductor material is silicon, the second semiconductor material is silicon germanium, and the dielectric feature is a silicon oxide feature.
8. The semiconductor structure of claim 5, further comprising a shallow trench isolation (STI) feature formed in the semiconductor substrate and surrounding the first active region, wherein the STI feature has a sidewall directly contacting a sidewall of the dielectric feature.
9. The semiconductor structure of claim 8, wherein the STI feature has a top surface being coplanar

with a top surface of the dielectric feature and a top of the semiconductor layer; the semiconductor layer includes a bottom surface being coplanar with a bottom surface of the dielectric feature; and the first active region continuously extends to the dielectric feature within the first area.

10. The semiconductor structure of claim 1, further comprising a first gate structure disposed within the first area and a second gate structure disposed within the second area, wherein the first active region includes a single channel, the first gate structure disposed around the single channel, and the second gate structure disposed around at least one of the plurality of channels.

11. A semiconductor structure, comprising: a semiconductor substrate having a first area and a second area; a first mesa of a first semiconductor material disposed within the first area, oriented in a first direction, and being isolated from the semiconductor substrate by a dielectric feature, wherein the first mesa has a $\langle 100 \rangle$ crystalline direction along the first direction; a second mesa of a second semiconductor material disposed within the second area and oriented in the first direction, wherein the second mesa has a $\langle 110 \rangle$ crystalline direction along the first direction, wherein the second mesa includes a plurality of channels vertically stacked and spaced away from each other; a first field-effect transistor (FET) of a first type conductivity and formed on the first mesa; a second FET of a second type conductivity and formed on the second mesa, wherein the second type conductivity is opposite to the first type conductivity; and a semiconductor layer of the first semiconductor material underlying the second mesa.

12. The semiconductor structure of claim 11, wherein the first FET is an n-type field-effect transistor (nFET); the second FET is a p-type field-effect transistor (pFET); the first semiconductor material is silicon; and the second semiconductor material is silicon germanium.

13. The semiconductor structure of claim 11, wherein each of the first mesa and the second mesa is in a crystalline structure with a top surface on a (100) crystal plane.

14. The semiconductor structure of claim 13, wherein the semiconductor substrate is a silicon substrate having a top surface on a (100) crystal plane; and the semiconductor substrate has a crystalline orientation $\langle 110 \rangle$ along the first direction.

15. The semiconductor structure of claim 11, wherein the first mesa continuously extends from a top surface of the first mesa to the dielectric feature; the second mesa is spaced away from the first mesa along a second direction that is orthogonal to the first direction; the dielectric feature spans between opposite edges of the first mesa along the second direction; and the semiconductor layer spans between opposite edges of the second mesa along the second direction.

16. The semiconductor structure of claim 11, further comprising a shallow trench isolation (STI) feature formed in the semiconductor substrate and surrounding the first mesa, wherein the STI feature has a sidewall directly contacting a sidewall of the dielectric feature; and the STI feature has a top surface being coplanar with a top surface of the dielectric feature and a top of the semiconductor layer.

17. A semiconductor structure, comprising: a semiconductor substrate having a first area and a second area; a dielectric feature disposed on the semiconductor substrate; a first active region of a first semiconductor material disposed within the first area, longitudinally oriented in a first direction, and being isolated from the semiconductor substrate by the dielectric feature, wherein the first active region has a $\langle 100 \rangle$ crystalline direction along the first direction; a second active region of a second semiconductor material disposed on the semiconductor substrate within the second area and longitudinally oriented in the first direction, wherein the second active region has a $\langle 110 \rangle$ crystalline direction along the first direction, wherein the second active region is directly landing on a semiconductor feature of the first semiconductor material, and wherein the second semiconductor material being different from the first semiconductor material in composition; and a shallow trench isolation (STI) feature formed in the semiconductor substrate, wherein the STI feature has a first sidewall contacting a sidewall of the dielectric feature and a second sidewall contacting a sidewall of the semiconductor feature, and wherein the second active region includes a plurality of channels vertically stacked and vertically spaced away from each other.

18. The semiconductor structure of claim 17, wherein the second active region is spaced away from the first active region along a second direction that is orthogonal to the first direction; the dielectric feature spans between opposite edges of the first active region along the second direction; and the semiconductor feature spans between opposite edges of the second active region along the second direction.

19. The semiconductor structure of claim 17, wherein each of the first and second active regions is in a crystalline structure with a top surface on a (100) crystal plane; the semiconductor substrate is a silicon substrate having a top surface on a (100) crystal plane; and the semiconductor substrate has a crystalline orientation $\langle 110 \rangle$ along the first direction.

20. The semiconductor structure of claim 17, further comprising an n-type field-effect transistor (nFET) formed on the first active region within the first area; and a p-type field-effect transistor (pFET) formed on the second active region within the second area, wherein the first semiconductor material is silicon; the second semiconductor material is silicon germanium; and the dielectric feature is a silicon oxide feature.
